DVD-ROM Drives: We used Nero DVDSpeed which showed the seek time, CPU utilisation spin down time and the drive speed of the DVD-ROM drives. The VOB file extraction test was carried out using the software Flask MPEG and the time taken for this was logged. We also used SiSoft Sandra 2002 Pro for checking the DVD-ROM drive's sequential read, random read and average access time. The time taken for extracting the first, middle and last track from an Audio CD was noted using the software Exact Audio Copy. An assorted read test was carried out where a combination of file sizes ranging from 1 KB to 100 MB were read from the CD-RW and copied to the hard disk and the time was noted. The disk registration time (the time taken by the drive to be recognised by the OS) was also noted.

Warranty and support: We logged the period of warranty for the CD-Writer and DVD-ROM drives. We also noted the type of support provided by the different brands. Some have a policy of replacing the faulty drive within the warranty period, while others only repair the drive.

Value for money: Here we calculated the features and performance delivered and at what cost. The best drive is that which is not very expensive and at the same time delivers impressive performance and has a host of features.

Test Analysis

There's a wide range of optical drives available in the market today. A DVD-ROM drive serves the purpose if you basically need an optical drive that will allow you to watch movies and don't need to back up data. A CD-Writer on the other hand, is what you should be looking at if you wish to archive and swap data.

Judging by the devices in this comparison test, all DVD-ROM drives have low CPU utilisation while dumping data form a DVD to the hard disk. If you wish to compress a DVD movie into the DivX format, the VOB extraction test will give you a good indication of a drive's performance while doing so. A drive with a low CPU utilisation score and a low VOB file extraction score will thus be the best for converting those DVD movies into the DivX format. We also found that the data seek times for DVD-ROM drives are now on par with those of CD-ROM drives.

If you plan to back up a lot of data using your CD-Writer, the test that you should especially consider is the Assorted Write Speed test, since the data that you are most likely to archive will consist of small individual files and folders. Additionally, the CD-RW is fast emerging as a replacement for the floppies and Zip disks; it is now the *de facto* means of swapping data with your friends or colleagues. The performance of the Writers under the RW-Assorted Write Speed test will give you a good idea of their capability to copy data onto a CD-RW.

CD-Writers

A CD-Writer works out to be an ideal backup solution since media is readily and cheaply available. A CD-ROM drive with 24x speed is now ubiquitous; moreover it is also capable of reading a CD-RW medium. Thus, if you have a CD-Writer and use a CD-RW to back up or swap data, you can rest assured that you will be able access the same on almost any PC.

Writers now offer burning speeds ranging from 32x to 48x, which means that the time taken to back up essential data is drastically reduced from 20 minutes to under 4 minutes.

Taking Care of Your Optical Drive

- Never use poor quality detergents and sprays for cleaning. Clean the drives using a soft, dry fabric.
- Keep the tray clean. Use a duster for thorough cleaning.
- Use a UPS (Uninterrupted Power Supply) for your computer and the Writer. This will protect your system from any surge or dip in power.
- Do not push the tray to close it; always use the eject button instead.
- Never try to force the tray out, if it doesn't eject automatically. Use the pin provided with the drive to manually remove the tray.
- Use a good quality lens cleaner to keep the lens clean, as this is one of the most important parts of the drive.

Features

The CD-Writers we received fell in the 32x to 48x speed bracket. The ASUS CRW-4816A, Iomega CDRW5529INT-B and LITEON LTR-48125 W offered the capability to burn a CD at 48x. While the models from LG and Krypton ran at 32x speed, which is the entry level for CD-Writers today, five of the entries offered a write speed of 40x—these were the Benq 4012P, Krypton BCE-4012M, LITEON LTR-40125 S, Samsung SW204, and Sony CRX-195A1. The Yamaha CRW-F1 offered a rather un-conventional speed of 44x.

All the CD-Writers supported burn proof technology and we experienced no errors while writing CDs at the maximum supported speed. It basically ensures that data is fed to the CD-Writing application in a constant stream as an interruption in data flow during the writing process can cause errors.

Also important for error-free CDs is the buffer size—higher the amount of buffer, lower the chances of ending up with a coaster. The buffer is the memory on the CD-Writer. It temporarily stores the data it receives from the hard drive or CD before it is passed on to the writing head. This ensures that the Writer is constantly supplied with the data to burn on the media. The LG 8320B, Yamaha CRW-F1 and Samsung SW204 were the only Writers to have an 8 MB buffer; the rest had a 2 MB buffer.

All the CD-Writers had adequate front panel indicators, but the noteworthy ones were the ASUS CRW-4816A and the Krypton BCE-3212M—both had an extra Play/Forward button whereas the other Writers only had the Eject/Stop button. The CD-Writers from LITEON had two LED indicators, one each for Read and Write.

Any device that has moving components needs to have especially good build quality, else it will soon develop mechanical snags regardless of how feature-rich and good the device is otherwise. Most of the CD-Writers had a decent build quality, except for the Yamaha CRW-F1—its tray was a little loose. The tray of the Samsung SW204 had a good build quality.

If a CD-Writer can be placed vertically, it gives you more flexibility if desktop space is a matter of concern for you. Although it is not advisable to keep a CD-Writer in the vertical position while burning data, it's all right if the drive is being used as a CD-ROM reader. Though all the CD-Writers could be used vertically, only the Yamaha CRW-F1 had a locking mechanism to hold the media in this position. However, when testing this feature on